LAN FENG

Ph.D. Candidate@EPFL, supervised by Prof. Alexandre Alahi
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PROFESSIONAL SUMMARY

I am a PhD candidate at EPFL working on **data-centric methods for foundation models**, with emphasis on LLM agents, generative world models, and end-to-end autonomy. I develop principled data-selection and closed-loop-evaluation frameworks such as **TAROT** (targeted data selection via optimal transport, ICML 2025), **UniTraj** for cross-dataset trajectory generalization (ECCV 2024), and **RAP** (3D rasterization augmented planning, ICLR 2026). My work has achieved leading results, including **first place in the 2025 Waymo End-to-End Driving Challenge** and state-of-the-art on the nuScenes, NAVSIM and Bench2Drive benchmarks. I extend these ideas to LLM agents in **Weak-for-Strong** (COLM 2025, w/ Stanford). I also study human-robot interaction through **SynH2R** and open-source generative simulation through **ScenarioNet** and **TrafficGen**. I will join **NVIDIA Research** in summer 2026 to work on world models and end-to-end driving.

EDUCATION

Ph.D. in Robotics, Control and Intelligent Systems, EPFL 2024 - 2028
Supervisor: Prof. Alexandre Alahi

M.S. in Robotics, Systems, and Control, ETH Zurich 2021 - 2023
Thesis: SynH2R: Synthesizing Hand-Object Motions for Learning Human-to-Robot Handovers
Advisor: Otmar Hilliges

B.E. in Navigation Engineering, Wuhan University 2016 - 2020
Thesis: LSTM-based Adaptive Stride Length Estimation

AWARDS AND HONORS

China National Scholarship 2019
1st Place in Waymo Interaction Trajectory Prediction Challenge 2021
1st Place on nuScenes Prediction Leaderboard 2024
1st Place in Waymo Vision-based End-to-End Driving Challenge 2025

PUBLICATIONS

1. **Lan Feng**, Yang Gao, Eloi Zabloki, Quanyi Li, Wuyang Li, Sichao Liu, Matthieu Cord, Alexandre Alahi "RAP: 3D Rasterization Augmented End-to-End Planning" *International Conference on Learning Representations* (ICLR 2026).
2. Fan Nie, **Lan Feng**, Haotian Ye, Weixin Liang, Pan Lu, Huaxiu Yao, Alexandre Alahi, James Zou "Weak-for-Strong: Training Weak Meta-Agent to Harness Strong Executors" (COLM 2025)
3. **Lan Feng***, Fan Nie*, Yuejiang Liu, and Alexandre Alahi. "TAROT: Targeted Data Selection via Optimal Transport" (ICML 2025).
4. **Lan Feng***, Mohammadhossein Bahari*, Kaouther Messaoud Ben Amor, Éloi Zablocki, Matthieu Cord, and Alexandre Alahi. "UniTraj: A Unified Framework for Scalable Vehicle Trajectory Prediction." *European Conference on Computer Vision* (ECCV 24).
5. **Lan Feng***, Sammy Christen*, Wei Yang, Yu-Wei Chao, Otmar Hilliges, and Jie Song. "SynH2R: Synthesizing Hand-Object Motions for Learning Human-to-Robot Handovers." *IEEE International Conference on Robotics and Automation* (ICRA 24).

6. **Lan Feng***, Quanyi Li*, Zhenghao Peng*, Zhizheng Liu, Chenda Duan, Wenjie Mo, and Bolei Zhou. "ScenarioNet: Open-Source Platform for Large-Scale Traffic Scenario Simulation and Modeling." *Neural Information Processing Systems, Dataset & Benchmark Track* (NeurIPS 23).

7. **Lan Feng***, Quanyi Li*, Zhenghao Peng*, Shuhan Liu, Bolei Zhou. "TrafficGen: Learning to Generate Diverse and Realistic Traffic Scenarios." *IEEE International Conference on Robotics and Automation* (ICRA 23).

8. **Lan Feng**, Sammy Christen, Jie Song. "Controllable Human Grasp Generation." *European Conference on Computer Vision* (ECCV 22 workshop).

9. Quanyi Li*, Zhenghao Peng*, **Lan Feng**, Zhenghai Xue, Qihang Zhang, Bolei Zhou. "MetaDrive: Composing Diverse Driving Scenarios for Generalizable Reinforcement Learning." *IEEE Transactions on Pattern Analysis and Machine Intelligence* (TPAMI 22).

10. Quanyi Li, Zhenghao Peng, Haibin Wu, **Lan Feng**, Bolei Zhou. "Human-AI Shared Control via Frequency-based Policy Dissection." *Advances in Neural Information Processing Systems* (NeurIPS 22).

11. **Lan Feng***, Qihang Zhang*, Yicheng Liu, Fan Li, Gang Sun, Chunxiao Liu, Bolei Zhou. "IP-MMT: Interaction Prediction via MultiModal Transformer." *Computer Vision and Pattern Recognition Conference Workshop* (CVPR workshop 21).

(* indicates joint first authors)

WORK EXPERIENCES

NVIDIA Research, Zurich Jun 2026 - Sep 2026
Research Intern (upcoming)

– World models and end-to-end driving.

Advanced Interactive Technologies Lab (AIT), ETH Zurich Feb 2022 - Sep 2023
Research Assistant (Supervisor: Otmar Hilliges)

– developed a generalizable RL-based robotic dexterous grasp algorithm.

Zhou Lab, UCLA Nov 2020 - Aug 2021
Research Assistant (Supervisor: Bolei Zhou)

– worked on the development of MetaDriverse, with a focus of data-driven traffic simulation.

– worked with Quanyi Li, Shuhan Tan, Zhenghao Peng and Bolei Zhou

SenseTime Nov 2020 - Aug 2021
Trainee Researcher

– worked on reinforcement learning-based traffic simulation and motion prediction.

– worked with Chunxiao Liu and Bolei Zhou

TECHNICAL SKILLS

Programming Languages	Python, LaTeX
Deep Learning	PyTorch, PyTorch-Lightning, Hugging Face, WandB, RLlib
LLM / Agents	OpenAI / Anthropic APIs
Robotics / Simulation	Isaac Gym, RaiSim, MetaDrive
Development Tools	Git, Docker, VS Code, Slurm